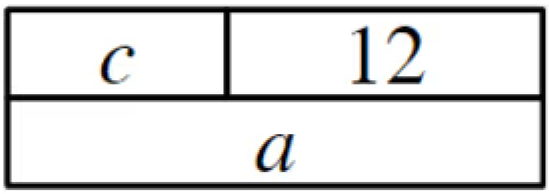
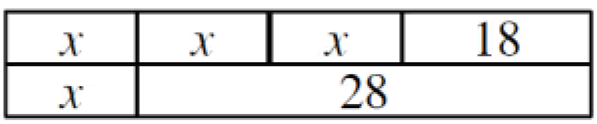
Equality in an equation

1 Which equations are represented by this bar model? Tick **2** correct answers



- [ ] $a + c = 12$
- [x] $a - c = 12$
- [ ] $c - a = 12$
- [ ] $12 + a = c$
- [x] $c = a - 12$

2 Which equations are represented by this bar model? Tick **2** correct answers

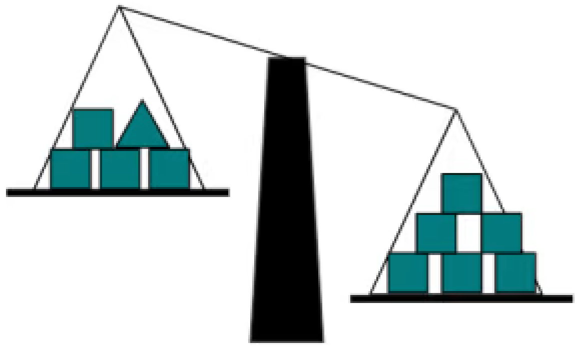


- [ ] $3x + 18 = 28$
- [x] $3x + 18 - 28 = x$
- [ ] $x + 28 - 3x + 18 = 0$
- [x] $x = x + 28 - 18 - 2x$
- [ ] $2x = 28 - 18 - x$

3 If $31 - x = 17$, which of these equations are also true? Tick **2** correct answers

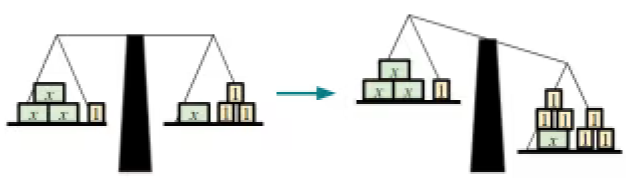
- [x] $31 - 17 = x$
- [ ] $17 - x = 31$
- [x] $x + 17 = 31$
- [ ] $x = 31 + 17$
- [ ] $x = 17 - 31$

4 These scales are not balanced. The mass of all the square objects are the same. Which statements are definitely true? Tick **1** correct answer



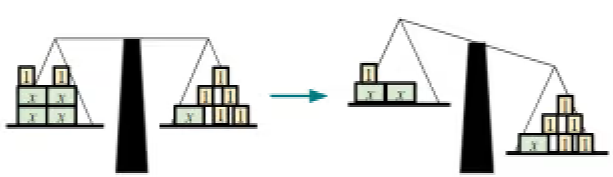
- [ ] The triangle is lighter than 1 square.
- [x] The triangle is lighter than 2 squares.
- [ ] The triangle has the same mass as 1 square.
- [ ] The triangle has the same mass as 2 squares.
- [ ] The triangle is heavier than 1 square.

5 The diagram shows a balanced scale on the left. When an operation is performed to the RHS, it is not balanced. What operation must be performed to the LHS to maintain equality? Tick **1** correct answer



- [ ] $+1$
- [x] $+3$
- [ ] $+\mathrm{x}$
- [ ] $\times 2$
- [ ] $\times 3$

6 The diagram shows a balanced scale on the left. When an operation is performed to the LHS, it is not balanced. What operation must be performed to the RHS to maintain equality? Tick **2** correct answers



- [ ] $+1$
- [ ] -1
- [ ] $+\mathrm{x}$ and -1
- [x] $-2\mathrm{x}$ and -1
- [x] $\times \frac{1}{2}$